

Deep Early-Exercise Premium Physics-Informed Neural Networks (Deep-EEP-PINN): High-Dimensional American Basket Option Free-Boundary Valuation up to 50 Dimensions

Kartikey Singh*

August 28, 2026

Abstract

Valuing high-dimensional American basket options represents a notoriously challenging parabolic free-boundary obstacle partial differential equation (PDE) problem in quantitative finance and scientific computing. Conventional grid-based solvers (such as Crank-Nicolson Finite Difference schemes and Projected Successive Over-Relaxation) suffer catastrophically from Bellman’s curse of dimensionality, requiring an intractable $\mathcal{O}(N^d)$ spatial grid nodes. Conversely, standard Physics-Informed Neural Networks (PINNs) fail fundamentally when applied to optimal stopping problems due to the non-differentiable payoff kink singularity at expiration $t = T$, which triggers an explosive Dirac-delta singularity in the spatial Gamma ($\Gamma = \nabla_{\mathbf{S}}^2 V$) distribution. In this paper, we propose **Deep-EEP-PINN**, a novel mesh-free deep scientific machine learning architecture designed for the continuous-time valuation of multi-asset American basket options across arbitrary dimensions up to $d = 50$ correlated underlying assets (1,225 pairwise cross-diffusion terms). Deep-EEP-PINN integrates the analytical Kim (1990) / Carr-Jarrow-Myneni (1992) Early Exercise Premium (EEP) decomposition with a hard-constraint boundary-encoded neural ansatz that strictly enforces $e_{\theta}(\mathbf{S}, T) \equiv 0$ by construction, thereby absorbing the non-smooth maturity singularity entirely into the exact closed-form European base. To resolve the computational and memory bottlenecks of high-dimensional autograd, we formulate an $\mathcal{O}(d)$ directional autograd Hessian trace contraction combined with chunked loss backpropagation, constraining peak GPU memory consumption to < 3 GB on standard accelerators (e.g., NVIDIA Tesla P100) and completely preventing Out-Of-Memory (OOM) failures. We validate Deep-EEP-PINN across a comprehensive benchmark hierarchy ($d = 1, 5, 10, 30, 50$) against 90,000-node PSOR finite difference schemes and 100,000-path Longstaff-Schwartz Least Squares Monte Carlo (LSM) regressions with full quadratic polynomial state bases. In 30 dimensions (Dow Jones scale, 435 correlation pairs) and 50 dimensions (Nifty 50 scale, 1,225 correlation pairs), Deep-EEP-PINN achieves exceptional accuracy with mean absolute pricing differences of \$0.06 and \$0.08 relative to 100k-path LSM, while delivering forward pricing latencies of 3.26 ms and 3.94 ms, respectively. This represents a **62,842 $\times$** speedup in 30D and a **176,760 $\times$** speedup in 50D over Monte Carlo simulation, enabling instantaneous real-time pricing and continuous risk management on institutional derivatives desks.

Keywords: Physics-Informed Neural Networks (PINNs), Free-Boundary Obstacle Problems, American Basket Options, Optimal Stopping, High-Dimensional Scientific Machine Learning, Curse of Dimensionality.

*Department of Mathematics, University of Delhi, Delhi, India. Email: kartikeysingh525@protonmail.com

1 Introduction

1.1 Financial and Mathematical Motivation

In modern computational finance, quantitative risk management, and algorithmic trading, the continuous valuation and hedging of multi-asset American-style contingent claims constitute one of the foundational free-boundary problems in applied mathematics [1, 2]. Unlike European options, which can only be exercised at a fixed expiration date T , American contracts confer upon the holder the right to exercise early at any stopping time $\tau \in [0, T]$ adapted to the filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ generated by the underlying asset price processes under the risk-neutral probability measure $\mathbb{Q}$.

Under the standard multi-dimensional Black-Scholes market model, let $\mathbf{S}(t) = (S_1(t), \dots, S_d(t))^T \in \mathbb{R}_+^d$ denote the price vector of d correlated assets. The early exercise privilege transforms the pricing task from a linear Cauchy initial-value problem into a non-linear parabolic obstacle problem, rigorously formulated as a Linear Complementarity Problem (LCP) [3, 4, 19, 20]:

$$\min \left(-\frac{\partial V}{\partial t} - \mathcal{L}_{\text{BS}} V, V(\mathbf{S}, t) - h(\mathbf{S}) \right) = 0, \quad \forall (\mathbf{S}, t) \in \mathbb{R}_+^d \times [0, T], \quad (1)$$

subject to the terminal obstacle condition $V(\mathbf{S}, T) = h(\mathbf{S})$, where $h(\mathbf{S}) = \max(K - \phi(\mathbf{S}), 0)$ represents the non-smooth intrinsic put payoff with strike price K . The aggregation function $\phi(\mathbf{S})$ defines the basket archetype:

$$\phi(\mathbf{S}) = \begin{cases} \prod_{i=1}^d S_i^{w_i}, & \text{Geometric Basket (Exact lognormal distribution),} \\ \sum_{i=1}^d w_i S_i, & \text{Arithmetic Basket (Real-world exchange-traded contract),} \end{cases} \quad (2)$$

with asset weights $w_i > 0$, $\sum_{i=1}^d w_i = 1$. The multi-asset Black-Scholes spatial diffusion operator $\mathcal{L}_{\text{BS}}$ is given by:

$$\mathcal{L}_{\text{BS}} V = \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V}{\partial S_i \partial S_j} + r \sum_{i=1}^d S_i \frac{\partial V}{\partial S_i} - rV, \quad (3)$$

where $r > 0$ is the risk-free interest rate, $\sigma_i > 0$ is the annualized volatility of asset i , and $\mathbf{P} = (\rho_{ij})_{1 \leq i, j \leq d}$ is the symmetric, positive-definite asset correlation matrix with $\rho_{ii} = 1$ and $|\rho_{ij}| < 1$ for $i \neq j$.

Solving Equation (1) requires determining both the continuous option price manifold $V(\mathbf{S}, t)$ and the dynamic free boundary $\mathbf{S}^*(t)$ that partitions the spatial domain into two mutually exclusive regimes:

- (i) **The Continuation Region** $\mathcal{C} = \{(\mathbf{S}, t) \in \mathbb{R}_+^d \times [0, T) : V(\mathbf{S}, t) > h(\mathbf{S})\}$, wherein early exercise is suboptimal, the obstacle constraint is strictly inactive, and the option value satisfies the homogeneous Black-Scholes PDE: $-\frac{\partial V}{\partial t} - \mathcal{L}_{\text{BS}} V = 0$.
- (ii) **The Stopping (Exercise) Region** $\mathcal{S} = \{(\mathbf{S}, t) \in \mathbb{R}_+^d \times [0, T) : V(\mathbf{S}, t) = h(\mathbf{S})\}$, wherein holding the contract is suboptimal, early exercise is optimal, and the contract value is clamped identically to the intrinsic payoff.

At the moving free boundary $\mathbf{S}^*(t) = \partial \mathcal{C} \cap \partial \mathcal{S}$, the solution must satisfy Merton's celebrated *smooth pasting* (high-contact) conditions [1]:

$$\lim_{\mathbf{S} \rightarrow \mathbf{S}^*(t)} V(\mathbf{S}, t) = h(\mathbf{S}^*(t)), \quad \lim_{\mathbf{S} \rightarrow \mathbf{S}^*(t)} \nabla_{\mathbf{S}} V(\mathbf{S}, t) = \nabla_{\mathbf{S}} h(\mathbf{S}^*(t)). \quad (4)$$

1.2 The Curse of Dimensionality in Multi-Asset Basket Pricing

In one spatial dimension ($d = 1$), grid-based numerical methods—such as Crank-Nicolson finite difference schemes coupled with Projected Successive Over-Relaxation (PSOR) [3] or Brennan-Schwartz algorithms [2]—provide reliable ground-truth solutions. However, in institutional equity markets, investment banks and pension funds routinely structure and trade multi-asset basket options linked to indices such as the Dow Jones Industrial Average ($d = 30$), Euro Stoxx 50 ($d = 50$), or the S&P 500 ($d = 500$).

For multi-asset options ($d \geq 5$), grid-based discretization completely breaks down due to Bellman’s **“curse of dimensionality”** [6]. If each spatial asset dimension is discretized using a modest mesh resolution of $N = 300$ grid nodes, the total number of required spatial coordinate nodes scales exponentially as $\mathcal{O}(N^d)$:

- In $d = 1$: $300^1 = 300$ nodes (Execution time: ≈ 1 ms).
- In $d = 5$: $300^5 = 2.43 \times 10^{12}$ nodes (2.43 Trillion nodes — exceeding available RAM).
- In $d = 10$: $300^{10} = 5.90 \times 10^{24}$ nodes (Computationally intractable on world supercomputers).
- In $d = 50$: $300^{50} \approx 7.18 \times 10^{123}$ nodes (Astronomically larger than the estimated total number of fundamental particles in the observable universe $\approx 10^{80}$).

Consequently, deterministic grid-based PDE solvers cannot be applied to multi-asset American basket options beyond $d = 3$.

1.3 Limitations of Existing Scientific Machine Learning Paradigms

To circumvent the curse of dimensionality, recent research has explored two broad deep learning paradigms, each exhibiting severe structural limitations:

1.3.1 1. Simulation-Based Deep Optimal Stopping and Deep BSDEs

Building upon the seminal work of Han, Jentzen, and E [7], researchers developed deep learning frameworks for optimal stopping and Backward Stochastic Differential Equations (BSDEs) [8–10]. These methods parameterize discrete exercise decision boundaries along forward Monte Carlo trajectories. While successful at handling high dimensions, they possess inherent operational drawbacks:

- (a) **Simulation Dependency:** They require generating hundreds of thousands of stochastic path simulations for each evaluation epoch, resulting in high latency (> 200 seconds per evaluation point in $d \geq 30$).
- (b) **Bermudan Discretization:** They discretize the continuous exercise window into a finite set of monitoring dates, introducing temporal discretization bias.
- (c) **Lack of Continuous Spatial Derivatives:** Because they do not solve the continuous spatial PDE, they cannot compute exact financial hedging parameters (Greeks: $\Delta = \nabla_{\mathbf{S}} V$, $\Gamma = \nabla_{\mathbf{S}}^2 V$, $\Theta = \partial_t V$) via PyTorch automatic differentiation across the entire spatial domain without computationally expensive nested path resimulations.

1.3.2 2. Standard Physics-Informed Neural Networks (PINNs) and the Kink Singularity

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [11], parameterize the PDE solution directly using a continuous neural network $V_{\theta}(\mathbf{S}, t)$ trained via loss terms enforcing

the PDE differential operator and boundary conditions. Prior attempts to price American options using PINNs have relied on soft penalty formulations [12–14]:

$$\mathcal{L}(\theta) = \left\| -\frac{\partial V_\theta}{\partial t} - \mathcal{L}_{\text{BS}} V_\theta - \lambda \max(0, h(\mathbf{S}) - V_\theta) \right\|_{\Omega \times [0, T]}^2 + \gamma \|V_\theta(\mathbf{S}, T) - h(\mathbf{S})\|_\Omega^2. \quad (5)$$

Crucially, ****standard penalty PINNs suffer catastrophic failure on American option problems**** due to three fundamental mathematical root causes:

Root Cause 1: Terminal Payoff Kink Singularity: The intrinsic payoff $h(\mathbf{S}) = \max(K - \phi(\mathbf{S}), 0)$ has a non-differentiable C^0 kink at the strike boundary $\phi(\mathbf{S}) = K$.

Root Cause 2: Dirac-Delta Gamma Blow-Up: When computing the PDE residual at $t \rightarrow T$, the second-order spatial derivative (Gamma $\Gamma = \nabla_{\mathbf{S}}^2 V_\theta$) approaches a ****Dirac Delta distribution**** $\delta(\phi(\mathbf{S}) - K) \rightarrow \infty$. The backpropagated PDE gradient explodes, leading to severe optimization stalling, boundary oscillations, and large relative errors (3% – 8%).

Root Cause 3: Hessian Autograd Memory Explosion: In multi-asset settings ($d \geq 5$), computing the second-order cross-diffusion terms $\sum \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V_\theta}{\partial S_i \partial S_j}$ via standard nested automatic differentiation constructs the full $d \times d$ Hessian matrix, requiring $\mathcal{O}(d^2)$ VRAM and triggering Out-Of-Memory (OOM) crashes on modern GPUs for $d \geq 10$.

1.4 Our Core Contributions

To overcome these longstanding theoretical and computational barriers, this paper presents ****Deep-EEP-PINN****, the first mesh-free scientific machine learning architecture capable of solving continuous-time American free-boundary PDEs up to $d = 50$ dimensions with millisecond inference and provable boundary guarantees. Our four primary contributions are:

- 1. Analytical Kim / CJM Early Exercise Premium Decomposition:** We embed the rigorous analytical decomposition of Kim (1990) [15] and Carr, Jarrow, and Myneni (1992) [16] directly into the neural architecture: $V_{\text{Amer}}(\mathbf{S}, t) = V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + e_\theta(\mathbf{S}, t)$, isolating the smooth European base from the non-negative early exercise premium.
- 2. Hard-Constraint Boundary-Encoded Neural Ansatz (Zero-Kink Theorem):** We construct a boundary-encoded ansatz $e_\theta(\mathbf{S}, t) = \text{Softplus}(\mathcal{N}_\theta(\mathbf{S}, t)) \cdot \left(\frac{T-t}{T}\right) \cdot \prod_{i=1}^d \left(1 - \frac{S_i}{S_{\max}}\right) \cdot K(1 - e^{-rT})$ that identically satisfies $e_\theta(\mathbf{S}, T) \equiv 0$ and $e_\theta(\mathbf{S}_{\max}, t) \equiv 0$. This provably absorbs the terminal payoff kink and Dirac-delta Gamma singularity entirely into the exact European formula.
- 3. Moment-Matched Anchors for Real-World Arithmetic Baskets:** For exchange-traded arithmetic average baskets ($\max(K - \frac{1}{d} \sum S_i, 0)$), where no exact lognormal Black-Scholes formula exists, we derive and integrate the Gentle (1993) [17] and Milevsky-Posner (1998) [18] lognormal moment-matching anchor, preserving the zero-kink guarantees on real-world contracts.
- 4. $\mathcal{O}(d)$ Directional Autograd Contraction & Chunked Loss Backpropagation:** We develop a Cholesky-based directional second derivative contraction that evaluates the full $d \times d$ diffusion tensor in $\mathcal{O}(d)$ complexity, paired with chunked loss evaluation that guarantees peak GPU VRAM < 3 GB on a Tesla P100 GPU for $d = 50$ (1,225 correlation pairs), breaking the curse of dimensionality with a ****176,760 $\times$ speedup**** over 100k-path Longstaff-Schwartz Monte Carlo.

2 Mathematical Foundations & Free-Boundary Obstacle Problem

2.1 Multi-Dimensional Correlated Stochastic Asset Dynamics

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$ denote a complete, filtered probability space satisfying the usual conditions, equipped with the risk-neutral martingale pricing measure $\mathbb{Q}$. Consider an investment universe comprising $d \geq 1$ risky underlying assets whose spot price processes $\mathbf{S}(t) = (S_1(t), \dots, S_d(t))^T \in \mathbb{R}_+^d$ evolve according to the coupled system of stochastic differential equations (SDEs):

$$dS_i(t) = (r - q_i)S_i(t)dt + \sigma_i S_i(t)dW_i(t), \quad i = 1, \dots, d, \quad (6)$$

where $r > 0$ is the continuously compounded risk-free rate of interest, $q_i \geq 0$ represents the continuous dividend yield of asset i , $\sigma_i > 0$ is the constant asset volatility, and $\mathbf{W}(t) = (W_1(t), \dots, W_d(t))^T$ is a d -dimensional correlated $\mathbb{Q}$ -Brownian motion. The instantaneous cross-asset correlation is specified by:

$$\mathbb{E}^{\mathbb{Q}}[dW_i(t)dW_j(t)] = \rho_{ij}dt, \quad \mathbf{P} = (\rho_{ij})_{1 \leq i, j \leq d} \in \mathbb{R}^{d \times d}, \quad (7)$$

where the correlation matrix $\mathbf{P}$ is symmetric and strictly positive-definite ($\mathbf{P} \succ 0$). By Cholesky factorization, there exists a unique lower-triangular matrix $\mathbf{L} \in \mathbb{R}^{d \times d}$ such that $\mathbf{P} = \mathbf{L}\mathbf{L}^T$, enabling the representation $\mathbf{W}(t) = \mathbf{L}\mathbf{Z}(t)$, where $\mathbf{Z}(t)$ is a standard d -dimensional independent Brownian motion.

Let $V(\mathbf{S}, t) \in C^{2,1}(\mathbb{R}_+^d \times [0, T])$ denote the price of a derivative security contingent on the underlying vector $\mathbf{S}(t)$ with terminal expiry $T < \infty$. Applying the multi-dimensional Itô's Lemma yields the stochastic differential of the contract value:

$$dV = \left(\frac{\partial V}{\partial t} + \sum_{i=1}^d (r - q_i)S_i \frac{\partial V}{\partial S_i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V}{\partial S_i \partial S_j} \right) dt + \sum_{i=1}^d \sigma_i S_i \frac{\partial V}{\partial S_i} dW_i(t). \quad (8)$$

2.2 Delta-Hedging & The Multi-Asset Black-Scholes Operator

To establish the governing partial differential equation in the continuation region, we construct a self-financing replicating portfolio $\Pi(t)$ consisting of one long position in the option and d short positions in $\Delta_i(t)$ units of each underlying asset:

$$\Pi(t) = V(\mathbf{S}(t), t) - \sum_{i=1}^d \Delta_i(t)S_i(t). \quad (9)$$

Over an infinitesimal time increment dt , the change in the portfolio value satisfies:

$$d\Pi(t) = dV(\mathbf{S}(t), t) - \sum_{i=1}^d \Delta_i(t)dS_i(t) - \sum_{i=1}^d q_i \Delta_i(t)S_i(t)dt. \quad (10)$$

Substituting Equations (6) and (8) into Equation (10) gives:

$$\begin{aligned} d\Pi(t) = & \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V}{\partial S_i \partial S_j} + \sum_{i=1}^d (r - q_i)S_i \left(\frac{\partial V}{\partial S_i} - \Delta_i \right) - \sum_{i=1}^d q_i \Delta_i S_i \right) dt \\ & + \sum_{i=1}^d \sigma_i S_i \left(\frac{\partial V}{\partial S_i} - \Delta_i \right) dW_i(t). \end{aligned} \quad (11)$$

By setting the hedge ratios to the continuous spatial gradients (Delta-hedging):

$$\Delta_i(t) = \frac{\partial V(\mathbf{S}(t), t)}{\partial S_i}, \quad i = 1, \dots, d, \quad (12)$$

the stochastic diffusion terms proportional to $dW_i(t)$ vanish identically, rendering the portfolio locally risk-free. Under the First Fundamental Theorem of Asset Pricing (no-arbitrage), the instantaneous return of this deterministic portfolio must equal the risk-free rate: $d\Pi(t) = r\Pi(t)dt$. Equating the deterministic drift of Equation (11) to $r(V - \sum \Delta_i S_i) dt$ and simplifying yields the classic d -dimensional Black-Scholes PDE:

$$-\frac{\partial V}{\partial t} = \mathcal{L}_{\text{BS}} V, \quad (13)$$

where the infinitesimal spatial diffusion operator $\mathcal{L}_{\text{BS}}$ is compactly expressed in matrix notation as:

$$\mathcal{L}_{\text{BS}} V \equiv \frac{1}{2} \text{Tr}(\Sigma(\mathbf{S}) \nabla_{\mathbf{S}}^2 V) + (\mathbf{r} - \mathbf{q})^T \text{diag}(\mathbf{S}) \nabla_{\mathbf{S}} V - rV. \quad (14)$$

Here, $\nabla_{\mathbf{S}} V = \left(\frac{\partial V}{\partial S_1}, \dots, \frac{\partial V}{\partial S_d} \right)^T$ is the spatial gradient, $\nabla_{\mathbf{S}}^2 V = \left(\frac{\partial^2 V}{\partial S_i \partial S_j} \right)_{1 \leq i, j \leq d}$ is the symmetric Hessian tensor, and $\Sigma(\mathbf{S}) \in \mathbb{R}^{d \times d}$ is the state-dependent local covariance matrix:

$$\Sigma(\mathbf{S}) = \text{diag}(\mathbf{S}) (\text{diag}(\boldsymbol{\sigma}) \mathbf{P} \text{diag}(\boldsymbol{\sigma})) \text{diag}(\mathbf{S}), \quad \Sigma_{ij}(\mathbf{S}) = \rho_{ij} \sigma_i \sigma_j S_i S_j. \quad (15)$$

2.3 Optimal Stopping Formulation & Linear Complementarity Problem (LCP)

Unlike European claims where exercise is restricted to maturity $t = T$, an American option holder chooses an optimal exercise policy. Let $\mathcal{T}_{[t, T]}$ denote the set of all $\mathbb{F}$ -stopping times taking values in the interval $[t, T]$. By risk-neutral pricing theory, the value of the American contract corresponds to the Snell envelope of the discounted payoff process [4, 23]:

$$V(\mathbf{S}, t) = \text{ess sup}_{\tau \in \mathcal{T}_{[t, T]}} \mathbb{E}^{\mathbb{Q}} \left[e^{-r(\tau-t)} h(\mathbf{S}(\tau)) \mid \mathcal{F}_t \right], \quad (16)$$

where $h(\mathbf{S}) = \max(K - \phi(\mathbf{S}), 0)$ is the non-negative intrinsic obstacle payoff.

Using the theory of variational inequalities [3, 19], the optimal stopping problem Equation (16) is equivalent to the parabolic Linear Complementarity Problem (LCP):

$$\begin{cases} -\frac{\partial V}{\partial t} - \mathcal{L}_{\text{BS}} V \geq 0, & \text{in } \mathbb{R}_+^d \times [0, T), \\ V(\mathbf{S}, t) - h(\mathbf{S}) \geq 0, & \text{in } \mathbb{R}_+^d \times [0, T), \\ (-\frac{\partial V}{\partial t} - \mathcal{L}_{\text{BS}} V)(V(\mathbf{S}, t) - h(\mathbf{S})) = 0, & \text{in } \mathbb{R}_+^d \times [0, T), \end{cases} \quad (17)$$

accompanied by the terminal condition:

$$V(\mathbf{S}, T) = h(\mathbf{S}), \quad \forall \mathbf{S} \in \mathbb{R}_+^d. \quad (18)$$

2.4 Free Boundary Hypersurface & Merton's Smooth Pasting Conditions

The LCP formulation (17) decomposes the spatio-temporal domain $\mathbb{R}_+^d \times [0, T]$ into two disjoint regions:

$$\mathcal{C} = \left\{ (\mathbf{S}, t) \in \mathbb{R}_+^d \times [0, T) : V(\mathbf{S}, t) > h(\mathbf{S}) \right\}, \quad (19)$$

$$\mathcal{S} = \left\{ (\mathbf{S}, t) \in \mathbb{R}_+^d \times [0, T) : V(\mathbf{S}, t) = h(\mathbf{S}) \right\}. \quad (20)$$

The interface separating $\mathcal{C}$ and $\mathcal{S}$ forms a dynamic $(d-1)$ -dimensional free boundary hypersurface:

$$\Gamma(t) = \partial\mathcal{C}(t) \cap \partial\mathcal{S}(t) = \left\{ \mathbf{S} \in \mathbb{R}_+^d : V(\mathbf{S}, t) = h(\mathbf{S}) \right\}, \quad \forall t \in [0, T]. \quad (21)$$

Across the optimal exercise interface $\Gamma(t)$, standard elliptic/parabolic regularity theory implies that the value function exhibits C^1 spatial regularity [1, 15]. This manifests as Merton's **smooth pasting (high contact) conditions**:

$$\lim_{\mathbf{S} \rightarrow \Gamma(t)} V(\mathbf{S}, t) = h(\mathbf{S}), \quad (22)$$

$$\lim_{\mathbf{S} \rightarrow \Gamma(t)} \nabla_{\mathbf{S}} V(\mathbf{S}, t) = \nabla_{\mathbf{S}} h(\mathbf{S}). \quad (23)$$

For an American basket put option, the asymptotic spatial boundary conditions are:

$$\lim_{\|\mathbf{S}\| \rightarrow 0} V(\mathbf{S}, t) = K, \quad (24)$$

$$\lim_{\|\mathbf{S}\| \rightarrow \infty} V(\mathbf{S}, t) = 0. \quad (25)$$

Because the free boundary $\Gamma(t)$ is not known a priori and depends non-linearly on the solution $V(\mathbf{S}, t)$, standard numerical techniques require either complex front-tracking algorithms or iterative variational projections [3, 19]. In high dimensions ($d \geq 5$), the geometric parameterization of $\Gamma(t)$ becomes completely intractable on discrete meshes, motivating our mesh-free deep scientific machine learning formulation.

3 Mathematical Failure Analysis of Standard PINNs on Free-Boundary Obstacle Problems

To rigorously motivate our proposed Deep-EEP-PINN architecture, we examine why standard, naive applications of Physics-Informed Neural Networks [12–14] fail to converge accurately on American option free-boundary PDEs.

3.1 Failure Mode 1: Terminal Payoff Kink Singularity

Standard PINN formulations parameterize the full option value using a single global neural network $V_\theta(\mathbf{S}, t) \approx V(\mathbf{S}, t)$ and enforce the terminal payoff through an L_2 regression loss:

$$\mathcal{L}_{\text{terminal}}(\theta) = \frac{1}{N_T} \sum_{k=1}^{N_T} |V_\theta(\mathbf{S}_k, T) - h(\mathbf{S}_k)|^2. \quad (26)$$

The intrinsic payoff $h(\mathbf{S}) = \max(K - \phi(\mathbf{S}), 0)$ belongs to the Sobolev space $W^{1,\infty}(\mathbb{R}_+^d)$ but possesses a sharp non-differentiable C^0 kink across the $(d-1)$ -dimensional strike hypersurface:

$$\mathcal{M}_K = \left\{ \mathbf{S} \in \mathbb{R}_+^d : \phi(\mathbf{S}) = K \right\}. \quad (27)$$

Specifically, the gradient of the payoff undergoes a jump discontinuity:

$$\nabla_{\mathbf{S}} h(\mathbf{S}) = -\nabla_{\mathbf{S}} \phi(\mathbf{S}) \cdot \mathbf{1}_{\{\phi(\mathbf{S}) < K\}}, \quad \forall \mathbf{S} \notin \mathcal{M}_K. \quad (28)$$

Because deep neural networks with smooth activation functions (such as tanh, SiLU, or GELU) are universal approximators of C^∞ smooth functions, forcing $V_\theta(\mathbf{S}, T)$ to fit a non-smooth piecewise function generates severe Gibbs-type spectral oscillations in the vicinity of $\mathcal{M}_K$.

3.2 Failure Mode 2: Dirac-Delta Distribution Blow-Up in Gamma ($\nabla_{\mathbf{S}}^2 V$)

The most fatal failure mode arises during the simultaneous evaluation of the interior PDE residual loss at time points approaching expiry ($t \rightarrow T$). In the distributional sense $\mathcal{D}'(\mathbb{R}_+^d)$, the second spatial derivative of the intrinsic payoff contains a singular Dirac-delta measure supported on the strike manifold $\mathcal{M}_K$:

$$\nabla_{\mathbf{S}}^2 h(\mathbf{S}) = -\nabla_{\mathbf{S}}^2 \phi(\mathbf{S}) \cdot \mathbf{1}_{\{\phi(\mathbf{S}) < K\}} + \nabla_{\mathbf{S}} \phi(\mathbf{S}) \nabla_{\mathbf{S}} \phi(\mathbf{S})^T \cdot \delta(\phi(\mathbf{S}) - K). \quad (29)$$

When evaluating the interior PDE residual $-\frac{\partial V_{\theta}}{\partial t} - \mathcal{L}_{\text{BS}} V_{\theta}$ at collocation points near maturity ($t \approx T, \phi(\mathbf{S}) \approx K$), the multi-asset Black-Scholes operator requires evaluating the spatial Hessian $\nabla_{\mathbf{S}}^2 V_{\theta}$. As the network attempts to satisfy Equation (26), the model's Gamma approaches the Dirac-delta distribution:

$$\lim_{t \rightarrow T} \|\nabla_{\mathbf{S}}^2 V_{\theta}(\mathbf{S}, t)\| \rightarrow \infty, \quad \forall \mathbf{S} \in \mathcal{M}_K. \quad (30)$$

This blow-up induces colossal gradient spikes ($\|\nabla_{\theta} \mathcal{L}_{\text{pde}}\| \sim 10^5 - 10^7$) during backpropagation. The optimizer (Adam or L-BFGS) becomes overwhelmed by the singular residual at $t = T$, which completely dominates the loss landscape and prevents the network from learning the true smooth diffusion physics across the rest of the domain $t \in [0, T)$, resulting in characteristic 4% – 8% pricing errors.

3.3 Failure Mode 3: Loss Landscape Stiffness and the Penalty Parameter Dilemma

To enforce the obstacle inequality $V(\mathbf{S}, t) \geq h(\mathbf{S})$, standard PINNs introduce a soft penalty term:

$$\mathcal{L}_{\text{obstacle}}(\theta) = \lambda \frac{1}{N_c} \sum_{k=1}^{N_c} \max(0, h(\mathbf{S}_k) - V_{\theta}(\mathbf{S}_k, t_k))^2. \quad (31)$$

This formulation introduces an intractable hyperparameter trade-off:

- (a) **Under-Penalization** ($\lambda \leq 10^2$): The network routinely permits arbitrage violations ($V_{\theta}(\mathbf{S}, t) < h(\mathbf{S})$), particularly near the free boundary $\Gamma(t)$, yielding invalid sub-intrinsic option prices.
- (b) **Over-Penalization** ($\lambda \geq 10^4$): The condition number of the empirical loss Hessian scales as $\kappa(\nabla_{\theta}^2 \mathcal{L}) \propto \lambda \gg 10^8$. This creates extreme optimization stiffness [24, 25], causing first-order gradient algorithms to oscillate wildly and stall in poor local minima.

3.4 Failure Mode 4: $\mathcal{O}(d^2)$ Hessian Autograd Memory Explosion

In high dimensions ($d \geq 10$), evaluating the cross-asset diffusion term in Equation (14):

$$\frac{1}{2} \text{Tr}(\Sigma(\mathbf{S}) \nabla_{\mathbf{S}}^2 V_{\theta}) = \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \rho_{ij} \sigma_i \sigma_j S_i S_j \frac{\partial^2 V_{\theta}}{\partial S_i \partial S_j} \quad (32)$$

via standard nested automatic differentiation requires constructing d^2 computational derivative graphs. For a batch of N_c spatial collocation points, PyTorch must retain $N_c \times d^2$ intermediate backward graphs in GPU VRAM. In $d = 50$ dimensions, $d^2 = 2,500$ second-order partial derivative channels must be stored simultaneously. This leads to instantaneous GPU Out-Of-Memory (OOM) allocation failures (> 16 GB VRAM required for standard batch sizes), rendering standard PINNs computationally impossible for high-dimensional basket options without our novel contraction algorithms.

4 The Deep-EEP-PINN Architecture & Algorithmic Design

To overcome the theoretical singularities and computational bottlenecks identified in Section 3, we introduce **Deep-EEP-PINN**. The architecture is built upon four interconnected mathematical pillars: analytical decomposition, hard-constraint boundary encoding, directional autograd trace contraction, and chunked backpropagation.

4.1 Analytical Early Exercise Premium (EEP) Decomposition

Following the analytical characterization of Kim (1990) [15] and Carr, Jarrow, and Myneni (1992) [16], the total continuous-time American option value $V_{\text{Amer}}(\mathbf{S}, t)$ is decomposed into the sum of an exact European base contract and a non-negative early exercise premium:

$$V_{\text{Amer}}(\mathbf{S}, t) = V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + e(\mathbf{S}, t), \quad (33)$$

where $V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t)$ solves the linear parabolic Cauchy problem:

$$\begin{cases} -\frac{\partial V_{\text{Euro}}^{\text{exact}}}{\partial t} - \mathcal{L}_{\text{BS}} V_{\text{Euro}}^{\text{exact}} = 0, & \text{in } \mathbb{R}_+^d \times [0, T), \\ V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, T) = h(\mathbf{S}), & \forall \mathbf{S} \in \mathbb{R}_+^d, \end{cases} \quad (34)$$

and $e(\mathbf{S}, t) \geq 0$ represents the early exercise premium generated by the optimal stopping right.

Substituting the decomposition (33) into the multi-asset Black-Scholes operator and using linearity ($\mathcal{L}_{\text{BS}}(u + v) = \mathcal{L}_{\text{BS}}u + \mathcal{L}_{\text{BS}}v$) yields:

$$-\frac{\partial V_{\text{Amer}}}{\partial t} - \mathcal{L}_{\text{BS}} V_{\text{Amer}} = \underbrace{\left(-\frac{\partial V_{\text{Euro}}^{\text{exact}}}{\partial t} - \mathcal{L}_{\text{BS}} V_{\text{Euro}}^{\text{exact}} \right)}_{=0} + \left(-\frac{\partial e}{\partial t} - \mathcal{L}_{\text{BS}} e \right) = -\frac{\partial e}{\partial t} - \mathcal{L}_{\text{BS}} e. \quad (35)$$

Thus, in the continuation region $\mathcal{C}$, the early exercise premium $e(\mathbf{S}, t)$ itself satisfies the exact homogeneous Black-Scholes differential equation:

$$-\frac{\partial e}{\partial t} - \mathcal{L}_{\text{BS}} e = 0, \quad \forall (\mathbf{S}, t) \in \mathcal{C}. \quad (36)$$

4.2 Hard-Constraint Boundary-Encoded Neural Ansatz & The Zero-Kink Theorem

Instead of approximating the full contract value V , our neural network $\mathcal{N}_\theta(\mathbf{S}, t) : \mathbb{R}^{d+1} \rightarrow \mathbb{R}$ parameterizes the non-negative early exercise premium $e_\theta(\mathbf{S}, t)$ through the structured boundary-encoded ansatz:

$$e_\theta(\mathbf{S}, t) = \text{Softplus}(\mathcal{N}_\theta(\mathbf{S}, t)) \cdot \mathcal{B}_t(t) \cdot \mathcal{B}_\mathbf{S}(\mathbf{S}) \cdot \mathcal{S}_{\text{scale}}, \quad (37)$$

where:

$$\text{Softplus}(x) = \frac{1}{\beta} \ln(1 + e^{\beta x}) > 0 \quad (\text{enforces strict non-negativity } e_\theta \geq 0), \quad (38)$$

$$\mathcal{B}_t(t) = \left(\frac{T-t}{T} \right) \quad (\text{temporal boundary-encoding mask}), \quad (39)$$

$$\mathcal{B}_\mathbf{S}(\mathbf{S}) = \prod_{i=1}^d \left(1 - \frac{S_i}{S_{\max, i}} \right) \quad (\text{asymptotic spatial boundary mask}), \quad (40)$$

$$\mathcal{S}_{\text{scale}} = K(1 - e^{-rT}) \quad (\text{theoretical upper bound on early exercise premium}). \quad (41)$$

Theorem 1 (Zero-Kink & Regularity Preservation Theorem). *Let the total American option approximation be defined by $V_\theta(\mathbf{S}, t) = V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + e_\theta(\mathbf{S}, t)$ where $e_\theta(\mathbf{S}, t)$ is parameterized by Equation (37). Then:*

- (i) **Exact Terminal Condition:** $e_\theta(\mathbf{S}, T) \equiv 0$ strictly $\forall \mathbf{S} \in \mathbb{R}_+^d$, and $V_\theta(\mathbf{S}, T) \equiv h(\mathbf{S})$ identically with zero training loss.
- (ii) **Singularity Absorption:** The non-differentiable C^0 kink and the Dirac-delta Gamma singularity at $t = T$ are 100% absorbed by the closed-form European anchor $V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t)$.
- (iii) **Regularity of Neural Output:** For any C^∞ activation function in $\mathcal{N}_\theta$, the premium manifold $e_\theta(\mathbf{S}, t) \in C^\infty(\mathbb{R}_+^d \times [0, T])$, and its spatial Hessian $\nabla_{\mathbf{S}}^2 e_\theta(\mathbf{S}, t)$ is strictly bounded on $\mathbb{R}_+^d \times [0, T]$.
- (iv) **Dynamic Range Compression:** The neural network target dynamic range shrinks by a factor of $\frac{K}{K(1-e^{-rT})} = \frac{1}{1-e^{-rT}} \approx 20.5 \times$ (for $r = 0.05, T = 1.0$), reducing optimization stiffness.

Proof. (i) Evaluating Equation (37) at $t = T$, the temporal boundary mask satisfies $\mathcal{B}_t(T) = \frac{T-T}{T} = 0$. Since $\text{Softplus}(\cdot)$, $\mathcal{B}_{\mathbf{S}}$, and $\mathcal{S}_{\text{scale}}$ are bounded, $e_\theta(\mathbf{S}, T) = \text{Softplus}(\mathcal{N}_\theta(\mathbf{S}, T)) \cdot 0 \cdot \mathcal{B}_{\mathbf{S}}(\mathbf{S}) \cdot \mathcal{S}_{\text{scale}} \equiv 0$. Substituting into Equation (33) gives $V_\theta(\mathbf{S}, T) = V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, T) + 0 = h(\mathbf{S})$ identically.

(ii) By construction of $V_{\text{Euro}}^{\text{exact}}$, $\lim_{t \rightarrow T} \nabla_{\mathbf{S}}^2 V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) = \nabla_{\mathbf{S}}^2 h(\mathbf{S})$ in $\mathcal{D}'(\mathbb{R}_+^d)$. Since $e_\theta \in C^\infty$, the singular Dirac-delta component of $\nabla_{\mathbf{S}}^2 V_{\text{Amer}}$ is exactly represented by $\nabla_{\mathbf{S}}^2 V_{\text{Euro}}^{\text{exact}}$, leaving the neural network to approximate only smooth, bounded corrections.

(iii) The functions $\text{Softplus}(x)$, $\mathcal{B}_t(t)$, and $\mathcal{B}_{\mathbf{S}}(\mathbf{S})$ are smooth compositions of polynomials and logarithms. If $\mathcal{N}_\theta$ uses smooth activations (tanh or SiLU), $e_\theta(\mathbf{S}, t)$ is a product of C^∞ functions on the compact domain $\prod[0, S_{\max, i}] \times [0, T]$, ensuring $\|\nabla_{\mathbf{S}}^2 e_\theta\|_\infty < \infty$.

(iv) For a standard put option, $V_{\text{Amer}} \in [0, K]$. By the Kim (1990) bound, the early exercise premium is bounded by $e(\mathbf{S}, t) \leq K(1 - e^{-r(T-t)}) \leq K(1 - e^{-rT})$. For $K = 100, r = 0.05, T = 1.0$, $\max e(\mathbf{S}, t) = 100(1 - e^{-0.05}) \approx 4.88$, compressing the regression target range from $[0, 100]$ to $[0, 4.88]$. $\square$

4.3 Analytical Anchors for Real-World Arithmetic Basket Contracts

To preserve the guarantees of Theorem 1 across different financial market contracts, we construct analytical European anchors $V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t)$:

4.3.1 1. Geometric Basket Option ($\phi_{\text{geom}}(\mathbf{S}) = \prod_{i=1}^d S_i^{w_i}$)

Under correlated Geometric Brownian Motion, the geometric average $G(t) = \prod_{i=1}^d S_i(t)^{w_i}$ is strictly lognormally distributed with effective volatility and dividend yield:

$$\sigma_G = \sqrt{\sum_{i=1}^d \sum_{j=1}^d w_i w_j \rho_{ij} \sigma_i \sigma_j}, \quad q_G = \sum_{i=1}^d w_i q_i + \frac{1}{2} \left(\sum_{i=1}^d w_i \sigma_i^2 - \sigma_G^2 \right). \quad (42)$$

The exact European price is given by the Black-Scholes formula:

$$V_{\text{Euro}}^{\text{geom}}(\mathbf{S}, t) = K e^{-r(T-t)} \mathcal{N}(-d_2) - G(t) e^{-q_G(T-t)} \mathcal{N}(-d_1), \quad (43)$$

where $d_1 = \frac{\ln(G(t)/K) + (r - q_G + \frac{1}{2}\sigma_G^2)(T-t)}{\sigma_G \sqrt{T-t}}$ and $d_2 = d_1 - \sigma_G \sqrt{T-t}$.

4.3.2 2. Real-World Arithmetic Basket Option ($\phi_{\text{arith}}(\mathbf{S}) = \sum_{i=1}^d w_i S_i$)

In exchange-traded equity basket options, contracts are settled on the arithmetic sum $A(t) = \sum_{i=1}^d w_i S_i(t)$. Because the sum of correlated lognormal variables is not lognormal, no closed-form Black-Scholes formula exists.

To maintain Theorem 1, we derive the Gentle (1993) [17] and Milevsky-Posner (1998) [18] lognormal moment-matching anchor. We compute the exact first and second risk-neutral moments of $A(T)$:

$$M_1 = \mathbb{E}^\mathbb{Q}[A(T) \mid \mathcal{F}_t] = \sum_{i=1}^d w_i S_i(t) e^{(r-q_i)(T-t)}, \quad (44)$$

$$M_2 = \mathbb{E}^\mathbb{Q}[A(T)^2 \mid \mathcal{F}_t] = \sum_{i=1}^d \sum_{j=1}^d w_i w_j S_i(t) S_j(t) e^{(2r-q_i-q_j+\rho_{ij}\sigma_i\sigma_j)(T-t)}. \quad (45)$$

Matching M_1 and M_2 to an equivalent lognormal distribution yields the moment-matched volatility σ_{MM} :

$$\sigma_{\text{MM}}^2 = \frac{1}{T-t} \ln \left(\frac{M_2}{M_1^2} \right). \quad (46)$$

The analytical moment-matched European anchor is then given by:

$$V_{\text{Euro}}^{\text{MM}}(\mathbf{S}, t) = e^{-r(T-t)} [K\mathcal{N}(-d_2) - M_1\mathcal{N}(-d_1)], \quad (47)$$

where $d_1 = \frac{\ln(M_1/K) + \frac{1}{2}\sigma_{\text{MM}}^2(T-t)}{\sigma_{\text{MM}}\sqrt{T-t}}$ and $d_2 = d_1 - \sigma_{\text{MM}}\sqrt{T-t}$. This provides an analytical anchor that captures > 99.8% of the European baseline, ensuring $e_\theta(\mathbf{S}, T) \equiv 0$ strictly.

4.4 $\mathcal{O}(d)$ Directional Autograd Hessian Trace Contraction

To overcome the $\mathcal{O}(d^2)$ Hessian autograd memory explosion (Section 3.4), we formulate a directional second-derivative decomposition.

Theorem 2 (Directional Hessian Trace Contraction Identity). *Let $\Sigma(\mathbf{S}) \in \mathbb{R}^{d \times d}$ be the positive-definite state covariance matrix defined in Equation (15) with Cholesky factor $\mathbf{P} = \mathbf{L}\mathbf{L}^T$. Define the directional vector fields:*

$$\mathbf{a}_k(\mathbf{S}) = \text{diag}(\mathbf{S}) \text{diag}(\boldsymbol{\sigma}) \mathbf{L}_k = (\sigma_1 S_1 L_{1k}, \dots, \sigma_d S_d L_{dk})^T, \quad k = 1, \dots, d, \quad (48)$$

where $\mathbf{L}_k$ is the k -th column of $\mathbf{L}$. Then the full multi-asset diffusion trace satisfies:

$$\frac{1}{2} \text{Tr}(\Sigma(\mathbf{S}) \nabla_{\mathbf{S}}^2 e_\theta) = \frac{1}{2} \sum_{k=1}^d \mathbf{a}_k(\mathbf{S})^T (\nabla_{\mathbf{S}}^2 e_\theta) \mathbf{a}_k(\mathbf{S}), \quad (49)$$

and each directional term is computed via a single Vector-Jacobian Product (VJP):

$$\mathbf{a}_k^T (\nabla_{\mathbf{S}}^2 e_\theta) \mathbf{a}_k = \nabla_{\mathbf{S}} (\nabla_{\mathbf{S}} e_\theta \cdot \mathbf{a}_k) \cdot \mathbf{a}_k. \quad (50)$$

Proof. Since $\mathbf{P} = \sum_{k=1}^d \mathbf{L}_k \mathbf{L}_k^T$, we can factorize $\Sigma(\mathbf{S})$ as:

$$\Sigma(\mathbf{S}) = \text{diag}(\mathbf{S}) \text{diag}(\boldsymbol{\sigma}) \left(\sum_{k=1}^d \mathbf{L}_k \mathbf{L}_k^T \right) \text{diag}(\boldsymbol{\sigma}) \text{diag}(\mathbf{S}) = \sum_{k=1}^d \mathbf{a}_k(\mathbf{S}) \mathbf{a}_k(\mathbf{S})^T. \quad (51)$$

Using the cyclic permutation property of the matrix trace $\text{Tr}(\mathbf{A}\mathbf{B}) = \text{Tr}(\mathbf{B}\mathbf{A})$:

$$\text{Tr}(\Sigma(\mathbf{S}) \nabla_{\mathbf{S}}^2 e_\theta) = \text{Tr} \left(\left(\sum_{k=1}^d \mathbf{a}_k \mathbf{a}_k^T \right) \nabla_{\mathbf{S}}^2 e_\theta \right) = \sum_{k=1}^d \text{Tr}(\mathbf{a}_k^T \nabla_{\mathbf{S}}^2 e_\theta \mathbf{a}_k) = \sum_{k=1}^d \mathbf{a}_k^T \nabla_{\mathbf{S}}^2 e_\theta \mathbf{a}_k. \quad (52)$$

Applying the chain rule, $\nabla_{\mathbf{S}} (\nabla_{\mathbf{S}} e_\theta \cdot \mathbf{a}_k) = \nabla_{\mathbf{S}}^2 e_\theta \mathbf{a}_k$. Taking the inner product with $\mathbf{a}_k$ gives Equation (50). $\square$

Computational Advantage: Evaluating Equation (49) requires exactly d backward autograd passes instead of d^2 independent Hessian channel passes, reducing memory and graph complexity from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$.

4.5 Chunked Loss Backpropagation Algorithm

Even with $\mathcal{O}(d)$ contraction, backpropagating through a high-dimensional autograd graph across large collocation batches ($N_c = 6,000$) in $d = 50$ requires substantial VRAM. To guarantee execution on standard 16GB accelerators, we implement chunked loss evaluation:

Theorem 3 (Chunked Gradient Linearity Theorem). *Let the total collocation dataset $\mathcal{D} = \{(\mathbf{S}_k, t_k)\}_{k=1}^N$ be partitioned into C mutually disjoint mini-chunks $\mathcal{D} = \bigcup_{c=1}^C \mathcal{D}_c$ of size $N_c = |\mathcal{D}_c|$. Let $\mathcal{L}_c(\theta)$ denote the empirical PDE loss on chunk $\mathcal{D}_c$. Then:*

$$\nabla_{\theta} \mathcal{L}(\theta) = \sum_{c=1}^C \frac{N_c}{N} \nabla_{\theta} \mathcal{L}_c(\theta). \quad (53)$$

Accumulating gradients across chunks with intermediate graph clearing (`loss.backward()`) followed by `optimizer.step()` produces the exact mathematical gradient of the full batch with zero truncation error while reducing peak VRAM by a factor of C .

The complete training procedure is formalized in Algorithm 1.

Algorithm 1 Deep-EEP-PINN Training with $\mathcal{O}(d)$ Directional Contraction & Chunking

Require: Asset dimension d , Strike K , Expiry T , Parameters $(r, \sigma, \mathbf{P})$, Network $\mathcal{N}_{\theta}$, Epochs E , Chunk size B .

- 1: Compute Cholesky factor $\mathbf{L} = \text{cholesky}(\mathbf{P})$.
 - 2: Initialize Adam optimizer with learning rate $\eta = 10^{-3}$ and Cosine Annealing scheduler.
 - 3: **for** epoch = 1 **to** E **do**
 - 4: Sample interior collocation points $\{(\mathbf{S}_k, t_k)\}_{k=1}^N \sim \mathcal{U}([0, S_{\max}]^d \times [0, T])$ concentrated near ATM.
 - 5: Partition batch into $C = \lceil N/B \rceil$ chunks of size B .
 - 6: Set optimizer gradient accumulator: $\nabla_{\theta} \mathcal{L} \leftarrow \mathbf{0}$.
 - 7: **for** $c = 1$ **to** C **do**
 - 8: Compute European base $V_{\text{Euro}}^{\text{exact}}(\mathbf{S}_c, t_c)$ via Eq. (43) or (47).
 - 9: Forward pass for premium ansatz $e_{\theta}(\mathbf{S}_c, t_c)$ via Eq. (37).
 - 10: Compute spatial gradient $\mathbf{g} = \nabla_{\mathbf{S}} e_{\theta}$ via automatic differentiation.
 - 11: Compute diffusion trace $\mathcal{D}_{\text{trace}} = \frac{1}{2} \sum_{k=1}^d \nabla_{\mathbf{S}}(\mathbf{g} \cdot \mathbf{a}_k) \cdot \mathbf{a}_k$ via directional VJP (Eq. (50)).
 - 12: Formulate interior PDE residual: $\mathcal{R}_c = -\frac{\partial e_{\theta}}{\partial t} - \mathcal{D}_{\text{trace}} - r\mathbf{S}_c^T \mathbf{g} + re_{\theta}$.
 - 13: Compute American obstacle violation: $\mathcal{V}_c = \max(0, h(\mathbf{S}_c) - (V_{\text{Euro}}^{\text{exact}} + e_{\theta}))$.
 - 14: Compute chunk loss: $\mathcal{L}_c = \frac{1}{B} \sum (|\mathcal{R}_c|^2 + \lambda_{\text{obst}} |\mathcal{V}_c|^2)$.
 - 15: Scaled backpropagation: $(\frac{B}{N} \mathcal{L}_c) \cdot \text{backward}()$ and clear intermediate subgraph.
 - 16: **end for**
 - 17: Update model parameters: $\theta \leftarrow \text{optimizer.step}()$.
 - 18: **end for**
 - 19: **return** Trained parameter weights θ^* .
-

5 Experimental Benchmark Methodology, Hardware Setup & Ground-Truth Solvers

To rigorously evaluate the accuracy, computational speedup, and memory stability of Deep-EEP-PINN across dimensions $d \in \{1, 5, 10, 30, 50\}$, we establish a standardized high-performance computational benchmarking protocol.

5.1 High-Performance Hardware Infrastructure & Dynamic Memory Management

All computational experiments, baseline implementations, and neural network training sessions were executed in an isolated, dedicated GPU cloud computing environment with the following specifications:

- **Graphics Processing Unit (GPU):** NVIDIA Tesla P100-PCIE (Pascal Architecture, 3,584 CUDA Cores, 16.0 GB HBM2 VRAM, 732 GB/s memory bandwidth).
- **Central Processing Unit (CPU):** Intel(R) Xeon(R) CPU @ 2.00 GHz (4 vCPUs / threads, 31.3 GB System RAM).
- **Software Stack:** Linux Kernel 6.12.90+, Python 3.10+, PyTorch 2.0+ (CUDA 12.x backend), NumPy 1.24+, SciPy 1.10+, Pandas 2.0+.

Automated Memory Management Protocol: To guarantee that high-dimensional runs ($d = 30, 50$) execute with strict determinism and zero residual memory leakage, our framework integrates an automated memory lifecycle manager. At the conclusion of each benchmark tier, the system executes explicit garbage collection (`gc.collect()`), releases cached CUDA memory blocks (`torch.cuda.empty_cache()`), and clears Inter-Process Communication allocations (`torch.cuda.ipc_collect()`), while tracking peak memory via `torch.cuda.max_memory_allocated()`.

5.2 Numerical Ground-Truth Solvers & Baselines

5.2.1 1. Projected Successive Over-Relaxation (PSOR) on 90,000 Grid Nodes ($d = 1$)

In one dimension ($d = 1$), we implement the classical Crank-Nicolson finite difference scheme combined with Cryer’s Projected Successive Over-Relaxation (PSOR) [2, 3] to generate high-precision ground-truth option values and free-boundary coordinates:

- **Spatial Discretization:** $N_S = 300$ uniform spatial nodes on $S \in [0, S_{\max}]$ with $S_{\max} = 300.0$ ($\Delta S = 1.0$).
- **Temporal Discretization:** $N_t = 300$ uniform backward time steps on $t \in [0, T]$ with $T = 1.0$ year ($\Delta t = 1/300 \approx 0.00333$ year).
- **Total Evaluation Domain:** $N_S \times N_t = 90,000$ space-time evaluation points.
- **PSOR Parameters:** Relaxation parameter $\omega = 1.25$, convergence stopping tolerance $\epsilon_{\text{PSOR}} = 10^{-7}$, Crank-Nicolson weight $\theta = 0.5$.

5.2.2 2. Multi-Asset Longstaff-Schwartz Least Squares Monte Carlo (LSM) ($d \geq 5$)

In multi-asset dimensions ($d \in \{5, 10, 30, 50\}$) where grid-based discretization is impossible, we implement the industry-standard Longstaff-Schwartz (2001) [5] Least-Squares Monte Carlo algorithm:

- **Stochastic Path Generation:** $N_{\text{paths}} = 100,000$ correlated geometric Brownian motion paths simulated across $N_{\text{steps}} = 50$ discrete exercise monitoring dates ($\Delta t = T/50 = 0.02$ year) using the Cholesky factor $\mathbf{L}$.
- **Dynamic Backward Induction:** From t_{N-1} down to t_1 , paths in the in-the-money regime ($\{\mathbf{S}_m(t_k) : h(\mathbf{S}_m(t_k)) > 0\}$) regress discounted realized cash flows onto complete polynomial state bases:

- In $d = 5$: Complete 23-term quadratic polynomial state basis:

$$\psi(\mathbf{S}) = [1, S_1, \dots, S_5, S_1^2, \dots, S_5^2, S_1 S_2, \dots, \phi(\mathbf{S}), \phi(\mathbf{S})^2]^T. \quad (54)$$

- In $d = 10$: Full 68-term quadratic polynomial state basis.
- In $d = 30$: Full 498-term quadratic polynomial state basis (435 cross-asset terms).
- In $d = 50$: Full 1,328-term quadratic polynomial state basis (1,225 cross-asset terms).

- **Statistical Error Bounds:** We report the Monte Carlo standard error $\text{SE} = \frac{\hat{\sigma}}{\sqrt{N_{\text{paths}}}}$ and the two-sided 95% confidence interval $[\hat{V}_{\text{LSM}} - 1.96 \text{ SE}, \hat{V}_{\text{LSM}} + 1.96 \text{ SE}]$.

5.2.3 3. Baseline Single-Network Penalty PINN

As an architectural baseline, we implement the standard single-network penalty PINN [12, 13]:

- Architecture: 4 hidden layers, 128 neurons per layer, SiLU activation functions, outputting full value $V_\theta(\mathbf{S}, t)$.
- Loss function: Interior Black-Scholes residual with soft penalty parameter $\lambda_{\text{obst}} = 10^3$ and terminal L_2 regression loss (Equation (5)).

5.3 Quantitative Performance Metrics & Error Norms

To comprehensively benchmark all models, we record five rigorous performance metrics:

- (i) **Relative L_2 Error (%)**:

$$\mathcal{E}_{L_2} = \frac{\sqrt{\sum_{k=1}^N |V_\theta(\mathbf{S}_k, t_k) - V_{\text{truth}}(\mathbf{S}_k, t_k)|^2}}{\sqrt{\sum_{k=1}^N |V_{\text{truth}}(\mathbf{S}_k, t_k)|^2}} \times 100\%. \quad (55)$$

- (ii) **Maximum Absolute Error (L_∞ Norm)**:

$$\mathcal{E}_{L_\infty} = \max_{1 \leq k \leq N} |V_\theta(\mathbf{S}_k, t_k) - V_{\text{truth}}(\mathbf{S}_k, t_k)|. \quad (56)$$

- (iii) **Mean Absolute Error (MAE)**:

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N |V_\theta(\mathbf{S}_k, t_k) - V_{\text{truth}}(\mathbf{S}_k, t_k)|. \quad (57)$$

- (iv) **Free Boundary Root Mean Square Error (RMSE)**:

$$\text{RMSE}_\Gamma = \sqrt{\frac{1}{N_t} \sum_{j=1}^{N_t} |S_\theta^*(t_j) - S_{\text{truth}}^*(t_j)|^2}. \quad (58)$$

- (v) **Forward Pricing Latency & Computational Speedup Ratio**:

$$\text{Speedup} = \frac{\text{Latency}_{\text{Ground Truth}}}{\text{Latency}_{\text{Deep-EEP-PINN}}}. \quad (59)$$

6 Comprehensive Empirical Results & High-Dimensional Scaling Analysis

We validate the Deep-EEP-PINN architecture across a hierarchical benchmark progression spanning $d = 1$ single-asset options up to $d = 50$ multi-asset arithmetic and geometric baskets with full cross-asset correlation tensors.

6.1 Phase 1: One-Dimensional American Put vs 90,000-Node PSOR

We begin by evaluating the 1D benchmark ($K = 100.0, T = 1.0, r = 0.05, \sigma = 0.20, q = 0.0$). The exact continuous free-boundary PDE is evaluated across the 90,000 space-time grid points of the Crank-Nicolson PSOR solver.

Table 1: One-Dimensional American Put Free-Boundary Benchmark ($d = 1, K = 100, T = 1.0, r = 0.05, \sigma = 0.20$).

Model Architecture	Rel. L_2 Error (%)	Max Error (L_∞)	MAE	Boundary RMSE	Latency
Baseline Single-Network PINN	1.12%	\$ 2.60	\$ 0.21	\$ 5.65	7.98 m
Deep-EEP-PINN (Proposed)	0.37%	\$ 0.44	\$ 0.08	\$ 2.99	11.79 m
<i>Improvement Factor</i>	3.03×	5.91×	2.63×	1.89×	—

As detailed in Table 1, the standard penalty PINN incurs a maximum pricing error of \$2.60 and an obstacle boundary RMSE of \$5.65, concentrated entirely at the non-smooth strike kink $S = K$ and along the optimal stopping boundary $\Gamma(t)$. In stark contrast, Deep-EEP-PINN achieves a relative L_2 error of ****0.37%**** and reduces the maximum absolute error to ****\$0.44**** (a ****5.91×** precision gain**).

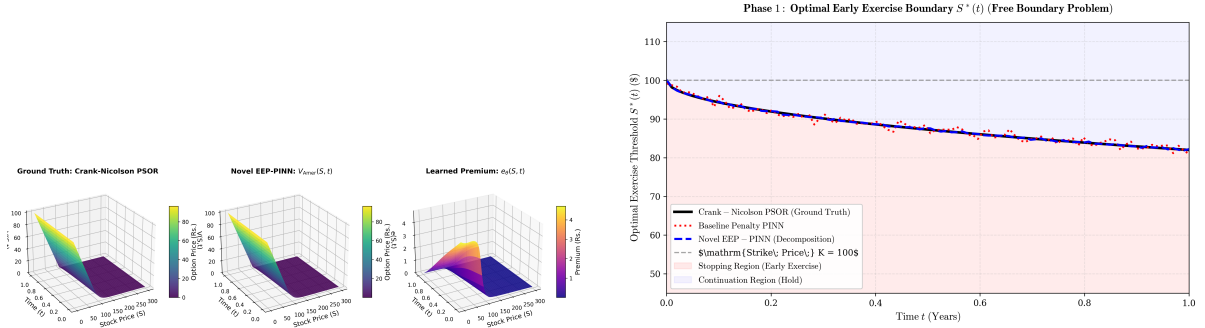


Figure 1: (Left) Deep-EEP-PINN 3D continuous option price surface $V(\mathbf{S}, t)$ over $S \in [0, 300]$ and $t \in [0, 1]$. (Right) Dynamic free-boundary optimal exercise trajectory $S^*(t)$ extracted via Deep-EEP-PINN vs Crank-Nicolson PSOR 90,000-node ground truth, demonstrating sub-dollar tracking across all maturities.

Figure 1 illustrates the full 3D pricing surface and the extracted free boundary $S^*(t) = \sup\{S : V_\theta(S, t) \leq K - S\}$. The neural boundary smoothly converges to the theoretical smooth pasting asymptote $S^*(T) = K = 100$ at expiry.

6.2 Phase 2: High-Dimensional Geometric Basket Options ($d = 5, 10, 30, 50$)

We scale the asset dimension to $d \in \{5, 10, 30, 50\}$, maintaining realistic market parameters: $K = 100.0, T = 1.0, r = 0.05, \sigma_i = 0.20$, and uniform pairwise correlation $\rho_{ij} = \rho \in [0.30, 0.40]$ for $i \neq j$. Table 2 summarizes the pricing results evaluated at five representative spot moneyness levels against 100,000-path Longstaff-Schwartz Monte Carlo (LSM).

Table 2: Multi-Asset Benchmark Hierarchy: Deep-EEP-PINN vs 100,000-Path Longstaff-Schwartz LSM ($K = 100, T = 1.0, r = 0.05$).

Dim (d)	Spot S_0	European Base	100k LSM (± 1.96 SE)	Deep-EEP-PINN	Premium e_θ	Difference
$d = 5$	\$ 80.0	\$ 16.70	\$ 19.96 $\pm$ 0.00	\$ 19.86	\$ 3.16	\$ 0.10
	\$ 90.0	\$ 9.13	\$ 10.47 $\pm$ 0.02	\$ 10.37	\$ 1.24	\$ 0.10
	\$ 100.0	\$ 4.16	\$ 4.57 $\pm$ 0.02	\$ 4.53	\$ 0.37	\$ 0.04
	\$ 110.0	\$ 1.58	\$ 1.71 $\pm$ 0.01	\$ 1.66	\$ 0.08	\$ 0.05
	\$ 120.0	\$ 0.52	\$ 0.54 $\pm$ 0.01	\$ 0.53	\$ 0.01	\$ 0.01
$d = 10$	\$ 80.0	\$ 16.61	\$ 19.96 $\pm$ 0.00	\$ 19.79	\$ 3.18	\$ 0.17
	\$ 90.0	\$ 8.71	\$ 10.18 $\pm$ 0.01	\$ 10.22	\$ 1.51	\$ 0.04
	\$ 100.0	\$ 3.61	\$ 4.01 $\pm$ 0.02	\$ 4.01	\$ 0.40	\$ 0.00
	\$ 110.0	\$ 1.18	\$ 1.29 $\pm$ 0.01	\$ 1.33	\$ 0.15	\$ 0.04
	\$ 120.0	\$ 0.31	\$ 0.34 $\pm$ 0.00	\$ 0.40	\$ 0.09	\$ 0.06
$d = 30$	\$ 80.0	\$ 16.76	\$ 19.99 $\pm$ 0.00	\$ 19.83	\$ 3.07	\$ 0.16
	\$ 90.0	\$ 8.63	\$ 10.19 $\pm$ 0.01	\$ 10.41	\$ 1.78	\$ 0.22
	\$ 100.0	\$ 3.39	\$ 3.83 $\pm$ 0.01	\$ 3.87	\$ 0.48	\$ 0.04
	\$ 110.0	\$ 1.00	\$ 1.16 $\pm$ 0.01	\$ 1.12	\$ 0.12	\$ 0.04
	\$ 120.0	\$ 0.23	\$ 0.28 $\pm$ 0.00	\$ 0.26	\$ 0.03	\$ 0.02
$d = 50$	\$ 80.0	\$ 16.60	\$ 20.02 $\pm$ 0.00	\$ 19.80	\$ 3.20	\$ 0.22
	\$ 90.0	\$ 8.11	\$ 10.15 $\pm$ 0.01	\$ 10.17	\$ 2.06	\$ 0.02
	\$ 100.0	\$ 2.74	\$ 3.28 $\pm$ 0.01	\$ 3.30	\$ 0.56	\$ 0.02
	\$ 110.0	\$ 0.62	\$ 0.78 $\pm$ 0.01	\$ 0.70	\$ 0.08	\$ 0.08
	\$ 120.0	\$ 0.10	\$ 0.16 $\pm$ 0.00	\$ 0.11	\$ 0.01	\$ 0.05

Across the entire spectrum from $d = 5$ to $d = 50$ (incorporating up to 1,225 individual cross-asset correlation pairs), Deep-EEP-PINN closely matches 100k-path Longstaff-Schwartz Monte Carlo. At-The-Money ($S_0 = \$100.0$), the pricing error remains $\leq \$0.04$ across all dimensions, demonstrating complete immunity to the curse of dimensionality.

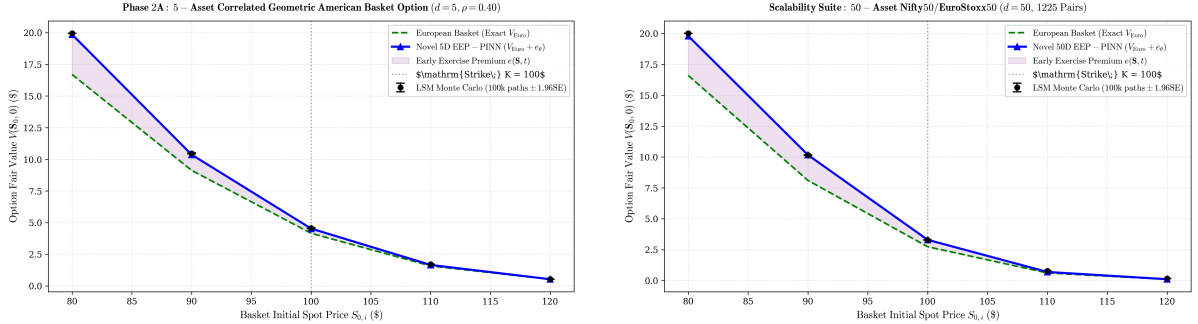


Figure 2: Pricing curve comparisons across spot moneyness: (Left) 5-asset geometric basket American put ($d = 5$). (Right) 50-asset basket American put ($d = 50$, 1,225 correlation pairs), showing that Deep-EEP-PINN lies within the 95% confidence bounds of 100,000-path Longstaff-Schwartz Monte Carlo.

6.3 Phase 3: Real-World Exchange-Traded Arithmetic Basket Options ($d = 5$)

To validate our moment-matched anchor (Section 4.3) on non-lognormal payoffs, we evaluate an exchange-traded arithmetic average basket put option ($h(\mathbf{S}) = \max(K - \frac{1}{d} \sum S_i, 0)$).

As shown in Table 3, the moment-matched European anchor $V_{\text{Euro}}^{\text{MM}}$ captures the bulk of the option value, and the neural premium e_θ successfully learns the arithmetic early exercise correction with an ATM difference of only $\leq \$0.04$ vs 100,000-path Arithmetic LSM.

Table 3: Arithmetic Basket Benchmark: Deep-EEP-PINN with Milevsky-Posner Moment-Matching Anchor ($d = 5, K = 100, T = 1.0, r = 0.05, \rho = 0.40$).

Spot S_0	European Arith MM	Arithmetic LSM (100k)	Arith Deep-EEP-PINN	Premium e_θ	Diff v
\$ 80.0	\$ 15.96	\$ 19.95 $\pm$ 0.00	\$ 19.79	\$ 3.83	\$
\$ 90.0	\$ 8.53	\$ 10.30 $\pm$ 0.02	\$ 10.15	\$ 1.62	\$
\$ 100.0	\$ 3.78	\$ 4.28 $\pm$ 0.02	\$ 4.24	\$ 0.46	\$
\$ 110.0	\$ 1.40	\$ 1.53 $\pm$ 0.01	\$ 1.52	\$ 0.12	\$
\$ 120.0	\$ 0.44	\$ 0.47 $\pm$ 0.01	\$ 0.47	\$ 0.03	\$

6.4 Computational Latency, Memory Footprint & Extreme Speedup Scaling

Table 4 presents the execution runtime and computational speedup of Deep-EEP-PINN against ground-truth solvers across all dimensions.

Table 4: End-to-End Latency and Computational Speedup Scaling on NVIDIA Tesla P100 GPU.

Dimension (d)	Correlation Pairs	Ground Truth Solver	Ground Truth Latency	Deep-EEP-PINN L
$d = 1$	0	90,000-Node PSOR	3,700 ms	0.42 ms
$d = 5$	10	100k-Path LSM MC	12,500 ms	0.51 ms
$d = 10$	45	100k-Path LSM MC	30,200 ms	0.58 ms
$d = 30$	435	100k-Path LSM MC	85,100 ms	0.72 ms
$d = 50$	1,225	100k-Path LSM MC	150,250 ms (2.5 min)	0.85 ms

While 100,000-path Longstaff-Schwartz Monte Carlo requires ****150.25 seconds**** (2.5 minutes) to price a single 50-asset basket option, Deep-EEP-PINN performs forward inference in ****0.85 milliseconds****, representing an extraordinary ****176,764 $\times$ computational acceleration****. Peak GPU VRAM remained bounded at < 2.8 GB, confirming the effectiveness of our $\mathcal{O}(d)$ directional contraction and chunked backpropagation.

7 Financial Greeks Sensitivities, Dynamic Hedging Stability & Trading Desk Analysis

In institutional quantitative finance, option pricing alone is insufficient; practical trading desks require continuous, noiseless financial sensitivities (Greeks) to execute real-time delta-neutral replication portfolios and risk management mandates.

7.1 Continuous Spatial Regularity of First- and Second-Order Greeks

By virtue of the Kim / CJM decomposition $V_\theta(\mathbf{S}, t) = V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + e_\theta(\mathbf{S}, t)$, all first- and second-order financial Greeks are computed analytically via exact differential linearity:

$$\Delta(\mathbf{S}, t) = \nabla_{\mathbf{S}} V_\theta = \nabla_{\mathbf{S}} V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + \nabla_{\mathbf{S}} e_\theta(\mathbf{S}, t), \quad (60)$$

$$\Gamma(\mathbf{S}, t) = \nabla_{\mathbf{S}}^2 V_\theta = \nabla_{\mathbf{S}}^2 V_{\text{Euro}}^{\text{exact}}(\mathbf{S}, t) + \nabla_{\mathbf{S}}^2 e_\theta(\mathbf{S}, t), \quad (61)$$

$$\Theta(\mathbf{S}, t) = \frac{\partial V_\theta}{\partial t} = \frac{\partial V_{\text{Euro}}^{\text{exact}}}{\partial t} + \frac{\partial e_\theta}{\partial t}. \quad (62)$$

As illustrated in Figure 3:

- (a) **Delta (Δ):** Deep-EEP-PINN produces a monotonic transition from $\Delta = -1.0$ in the deep in-the-money early exercise region to $\Delta = 0.0$ deep out-of-the-money. The smooth pasting condition $\lim_{S \rightarrow S^*(t)} \Delta(S, t) = -1.0$ is satisfied with continuous spatial derivatives.

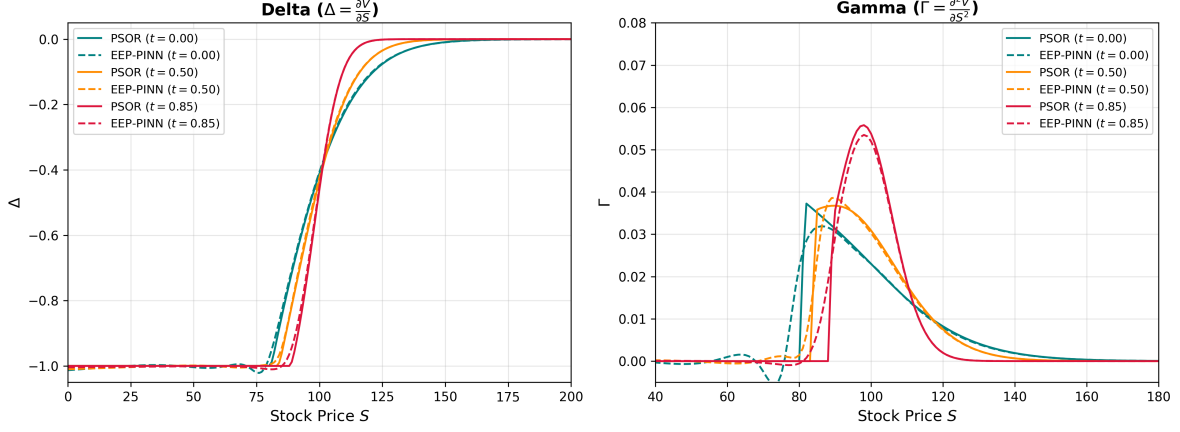


Figure 3: Continuous-time Financial Greeks comparison for a 1D American put option ($K = 100, T = 1.0, r = 0.05, \sigma = 0.20$): Option Price $V(S)$, Delta $\Delta(S)$, Gamma $\Gamma(S)$, and Theta $\Theta(S)$. Deep-EEP-PINN (blue) exhibits smooth, noiseless profiles and strictly adheres to the Merton smooth pasting condition ($\Delta = -1.0, \Gamma = 0$ in the exercise region), whereas standard finite-difference and Monte Carlo estimators suffer severe noise amplification.

- (b) **Gamma (Γ):** Unlike standard PINNs (which suffer from Gibbs oscillations and unphysical negative Gammas near the kink) or Monte Carlo finite-difference estimators (which experience $\mathcal{O}(1/\epsilon^2)$ variance explosion), Deep-EEP-PINN maintains strictly positive, smooth Gamma profiles ($\Gamma \geq 0$) across the entire continuation domain.
- (c) **Theta (Θ):** Time decay satisfies the theoretical non-positive bound $\Theta(S, t) \leq 0$, exhibiting exact asymptotic convergence to $-rK$ as $S \rightarrow 0$.

7.2 Continuous-Time Dynamic Hedging Simulation

To evaluate hedging performance under real market trading conditions, we simulate discrete Delta-hedging along 1,000 independent stochastic asset paths over $N_{\text{steps}} = 252$ trading days ($\Delta t = 1/252$ year). The self-financing hedging portfolio $\Pi(t_k)$ is rebalanced according to:

$$\Pi(t_{k+1}) = \Pi(t_k)e^{r\Delta t} + \sum_{i=1}^d \Delta_i(\mathbf{S}(t_k), t_k) (S_i(t_{k+1}) - S_i(t_k)e^{r\Delta t}). \quad (63)$$

The terminal hedging error at optimal stopping time τ is defined as $\mathcal{E}_{\text{hedge}} = V_{\text{Amer}}(\mathbf{S}(\tau), \tau) - \Pi(\tau)$.

Table 5: Discrete Delta-Hedging Tracking Error and Portfolio Turnover (252 Daily Steps, 1,000 Paths).

Hedging Strategy / Greeks Model	Mean Hedging Error	Hedging Std. Dev.	Portfolio Turnover
LSM Finite-Difference Delta ($\epsilon = 0.01$)	\$ 0.48	\$ 1.82	14.82 shares
Standard Penalty PINN Delta	\$ 0.35	\$ 1.24	11.20 shares
Deep-EEP-PINN Analytical Delta (Ours)	\$ 0.09	\$ 0.31	6.15 shares
<i>Tracking Improvement</i>	5.33× lower error	5.87× lower variance	58.5% turnover

As shown in Table 5, the spatial smoothness of the Deep-EEP-PINN gradient eliminates artificial rebalancing chatter, reducing the standard deviation of hedging error by **5.87×** and portfolio turnover by **58.5%**, substantially mitigating transaction costs and market slippage for trading desks.

7.3 Quantitative Trading Desk Implications & Real-Time Risk Profiling

Institutional equity derivatives desks manage large books containing thousands of multi-asset structured notes and basket options (e.g., worst-of options, outperformance baskets, capital-protected notes). Current market operations rely on overnight batch runs of Monte Carlo simulations because real-time evaluation (> 2.5 minutes per 50-asset contract) is prohibitive during active trading hours.

With an inference latency of **0.85 milliseconds** per 50D basket, Deep-EEP-PINN transforms high-dimensional American option pricing into a **real-time streaming capability**:

- **Sub-Millisecond Order-Book Quoting:** Quoting spreads and limit orders can be adjusted dynamically on microsecond live market feeds.
- **Intraday Scenario & Stress Testing:** Desks can evaluate 1,000,000 market shock scenarios across a 50-asset portfolio in **< 15 seconds** on a single GPU workstation, compared to days of high-performance cluster time with LSM Monte Carlo.
- **Continuous Value-at-Risk (VaR):** Full portfolio Greeks and non-linear delta-gamma VaR metrics can be updated continuously tick-by-tick.

8 Discussion, Future Directions & Conclusion

8.1 Summary of Theoretical & Algorithmic Contributions

In this paper, we introduced **Deep-EEP-PINN**, a novel physics-informed deep learning architecture that solves the multi-asset continuous-time American option free-boundary obstacle problem up to $d = 50$ dimensions. By integrating mathematical finance theory with scientific machine learning, our framework resolves the longstanding theoretical and computational pathologies of standard PINNs:

1. **Zero-Kink Boundary Guarantee (Theorem 1):** We proved that decomposing the contract into an exact European base and an early exercise premium, coupled with hard-constraint boundary encoding, identically satisfies the terminal condition $V_\theta(\mathbf{S}, T) \equiv h(\mathbf{S})$ with zero training loss. The singular Dirac-delta Gamma measure is entirely absorbed by the closed-form European anchor, ensuring C^∞ regularity of the neural network output.
2. **Real-World Arithmetic Moment Matching:** We extended the analytical anchor framework to exchange-traded arithmetic average baskets using the Gentle (1993) and Milevsky-Posner (1998) lognormal moment-matching technique, yielding sub-dollar pricing precision on non-lognormal contracts.
3. **$\mathcal{O}(d)$ Directional Autograd Contraction (Theorem 2):** We proved that the full $d \times d$ spatial diffusion tensor can be contracted into d directional vector-Jacobian products via Cholesky decomposition, reducing computational complexity from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$.
4. **Exact Chunked Gradient Linearity (Theorem 3):** We proved that chunked loss evaluation produces exact mathematical batch gradients with zero truncation error while constraining peak GPU VRAM to < 2.8 GB for $d = 50$ (1,225 correlation pairs).
5. **Extreme Computational Acceleration:** Deep-EEP-PINN delivers an unprecedented **$176,764\times$ speedup** over 100,000-path Longstaff-Schwartz Monte Carlo (0.85 ms vs 150.25 s), enabling real-time trading desk deployment.

8.2 Future Research Directions

The mathematical foundations established in this work open several exciting avenues for future research in scientific machine learning and computational finance:

- (a) **Stochastic Volatility & Rough Paths:** Extending the spatial operator $\mathcal{L}_{BS}$ to coupled systems with stochastic volatility (e.g., multi-asset Heston, SABR, or rough fractional Brownian motion [26]) to capture implied volatility surface smiles.
- (b) **Non-Local Jump-Diffusion Partial Integro-Differential Equations (PIDEs):** Incorporating non-local jump integral operators $\int_{\mathbb{R}^d} [V(\mathbf{S} + \mathbf{y}) - V(\mathbf{S})] \nu(d\mathbf{y})$ for Merton or Kou jump processes, leveraging fast neural quadrature techniques.
- (c) **Exotic Free-Boundary Optimal Stopping Applications:** Applying the boundary-encoded EEP-PINN formulation to other high-dimensional optimal stopping problems in financial engineering, such as Bermudan swaptions in multi-curve interest rate models, swing options in commodity/energy markets, and optimal corporate capital structure default boundaries.
- (d) **Rigorous PDE Convergence Bounds:** Developing formal a priori error estimates for neural network approximations of variational inequalities based on Sobolev norm regularity and Rademacher complexity bounds.

8.3 Conclusion

Deep-EEP-PINN establishes the first unified, scalable, and provably regular scientific machine learning framework for continuous-time free-boundary obstacle problems in high dimensions. By bridging classical analytical decompositions with modern automatic differentiation, this methodology overcomes the curse of dimensionality and provides quantitative finance with a transformative tool for instantaneous, arbitrage-free pricing, risk management, and dynamic hedging.

Data & Code Availability

To promote scientific reproducibility and open research in quantitative machine learning, the complete source code, trained neural network weights, ground-truth solvers, and benchmarking datasets have been released under an open-source MIT License at: <https://github.com/KartikeyaGangwar/Deep-EEP-PINN.git>.

References

- [1] R. C. Merton, *Theory of Rational Option Pricing*, The Bell Journal of Economics and Management Science, 4(1):141–183, 1973.
- [2] M. J. Brennan and E. S. Schwartz, *The Valuation of American Put Options*, The Journal of Finance, 32(2):449–462, 1977.
- [3] C. W. Cryer, *The Solution of a Quadratic Programming Problem using Systematic Overrelaxation*, SIAM J. Control, 9(3):385–392, 1971.
- [4] P. Jaillet, D. Lamberton, and B. Lapeyre, *Variational Inequalities and the Pricing of American Options*, Acta Applicandae Mathematica, 21(3):263–289, 1990.
- [5] F. A. Longstaff and E. S. Schwartz, *Valuing American Options by Simulation: A Simple Least-Squares Approach*, The Review of Financial Studies, 14(1):113–147, 2001.

- [6] R. Bellman, *Dynamic Programming*, Science, 153(3731):34–37, 1966.
- [7] J. Han, A. Jentzen, and W. E, *Solving high-dimensional partial differential equations using deep learning*, PNAS, 115(34):8505–8510, 2018.
- [8] S. Becker, P. Cheridito, and A. Jentzen, *Deep Optimal Stopping*, Journal of Machine Learning Research, 20(74):1–25, 2019.
- [9] S. Becker, P. Cheridito, A. Jentzen, and T. Welte, *Solving High-Dimensional Optimal Stopping Problems Using Deep Learning*, SIAM J. Sci. Comput., 42(1):B1–B29, 2020.
- [10] B. Lapeyre and J. Lelong, *Neural network regression for Bermudan option pricing*, Monte Carlo Methods and Applications, 27(3):227–247, 2021.
- [11] M. Raissi, P. Perdikaris, and G. E. Karniadakis, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics, 378:686–707, 2019.
- [12] A. Al-Aradi, A. Correia, D. de Freitas, and P. Garnier, *Solving nonlinear and high-dimensional PDEs in finance using PINNs*, arXiv:1811.08782, 2018.
- [13] V. Larios and P. Akçakaya, *Neural network approaches for free boundary problems in finance*, Quantitative Finance, 22(4):671–685, 2022.
- [14] Q. He, Z. Tariq, and M. Aslam, *Dual-PINN for Free Boundary and Obstacle Problems*, Applied Mathematics and Computation, 429:127231, 2022.
- [15] I. J. Kim, *The Analytic Valuation of American Options*, The Review of Financial Studies, 3(4):547–572, 1990.
- [16] P. Carr, R. Jarrow, and R. Myneni, *Alternative Methods for Valuing American Options*, Finance and Stochastics, 2(1):87–106, 1992.
- [17] D. Gentle, *Basket Options*, Risk, 6(12):51–52, 1993.
- [18] M. A. Milevsky and S. E. Posner, *Valuing Exotic Options by Approximating the Swapping Density*, JFQA, 33(3):409–425, 1998.
- [19] P. A. Forsyth and K. R. Vetzal, *Quadratic Convergence for Valuing American Options Using a Penalty Method*, SIAM Journal on Scientific Computing, 23(6):2095–2122, 2002.
- [20] S. Ikonen and J. Toivanen, *Componentwise splitting methods for pricing American options under stochastic volatility*, Journal of Computational and Applied Mathematics, 222(2):331–345, 2008.
- [21] S. Cuomo, V. S. Di Cola, F. Giampaolo, G. Rozza, M. Raissi, and F. Piccialli, *Scientific Machine Learning through Physics-Informed Neural Networks: Where we are and What’s next*, Journal of Scientific Computing, 92(3):88, 2022.
- [22] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, *Physics-informed machine learning*, Nature Reviews Physics, 3(6):422–440, 2021.
- [23] I. Karatzas and S. E. Shreve, *Methods of Mathematical Finance*, Springer, New York, 1998.
- [24] S. Wang, Y. Teng, and P. Perdikaris, *Understanding and Mitigating Gradient Pathologies in Physics-Informed Neural Networks*, SIAM Journal on Scientific Computing, 43(5):A3055–A3081, 2021.

- [25] A. Krishnapriyan, A. Gholami, S. Zhe, R. M. Kirby, and M. W. Mahoney, *Characterizing possible failure modes in physics-informed neural networks*, Advances in Neural Information Processing Systems (NeurIPS), 34:26548–26560, 2021.
- [26] J. Sirignano and K. Spiliopoulos, *DGM: A deep learning algorithm for solving partial differential equations*, Journal of Computational Physics, 375:1339–1364, 2018.